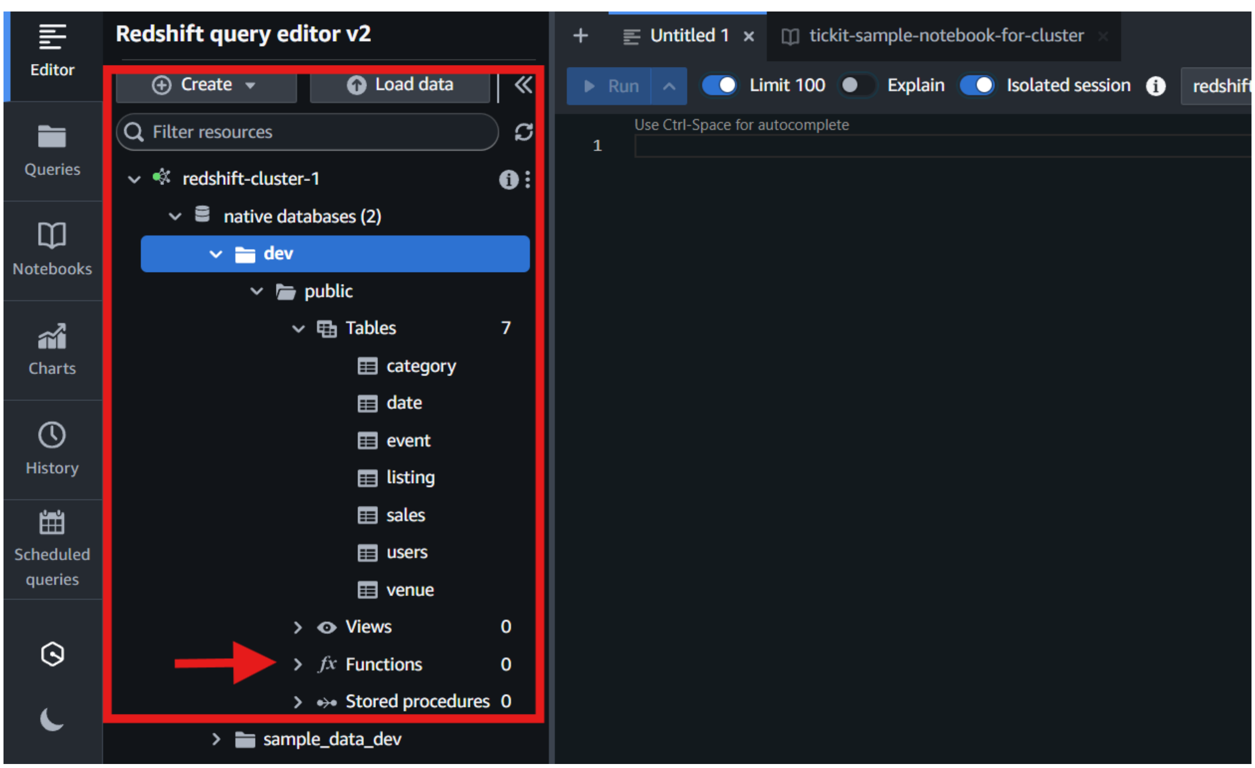
User Defined Function (UDF) in Amazon Redshift

To Begin with the Lab

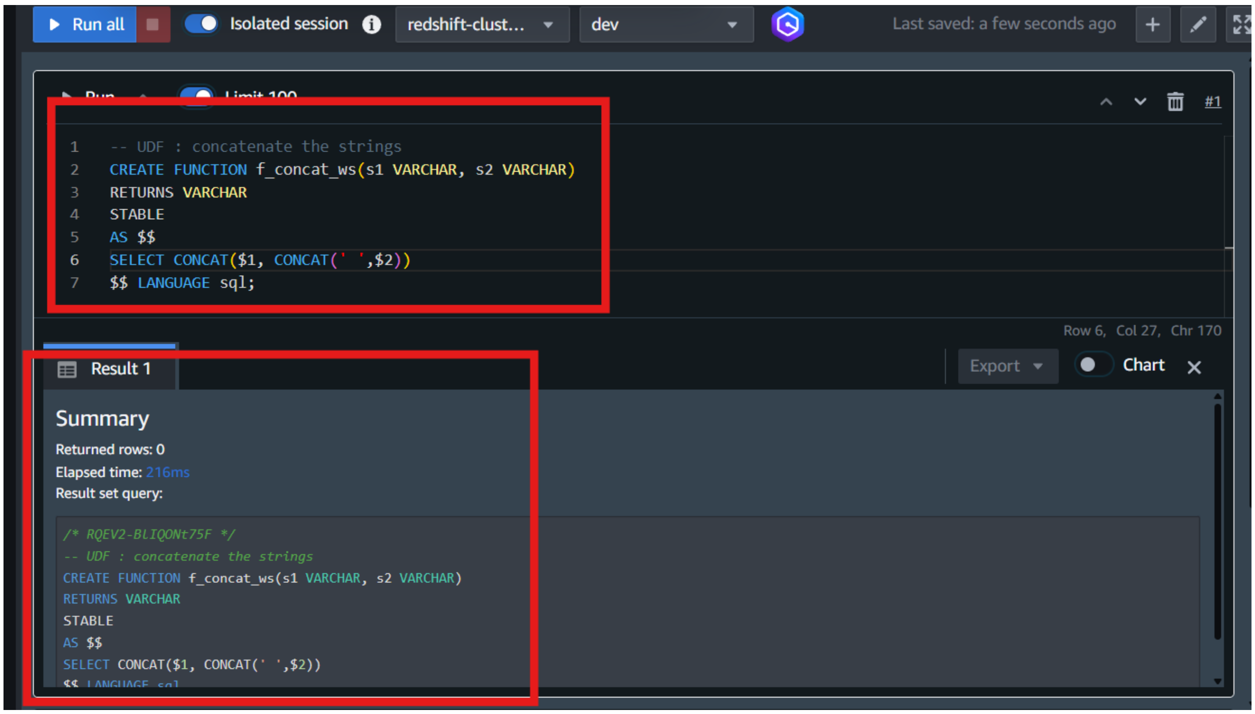
Summary of the Lab

In this lab, a SQL-based User Defined Function named f_concat_ws was created in **Amazon Redshift** to concatenate two strings with a space between them. The lab used SQL UDF functionality to simplify repetitive string formatting logic for analytics queries. After creating the function, the Functions section inside **Redshift Query Editor v2** was refreshed to verify that the custom UDF had been created successfully. The function was then executed against the users table to combine first names and last names into a single formatted output. Finally, the query results were validated successfully by confirming that usernames and full names were displayed correctly with spaces between them.

- Sign in to the AWS Management Console
- Navigate to **Amazon Redshift** → open **Redshift Query Editor v2**
- Connect to the Redshift cluster with the Tickit sample database loaded
- Select the dev database

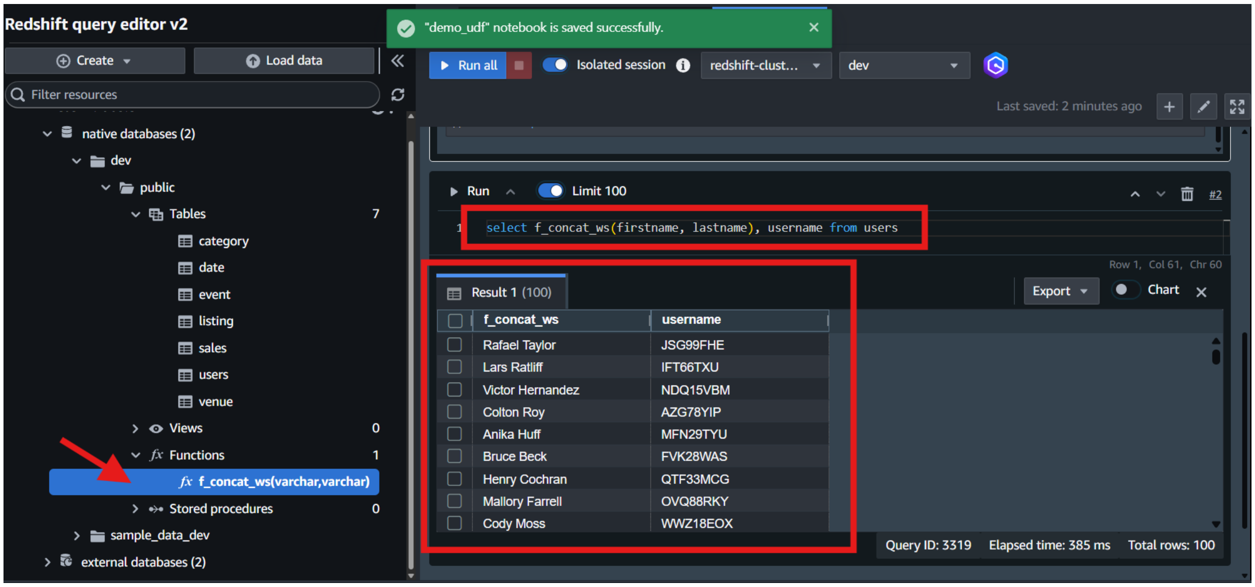


- Expand the Functions section from the left navigation panel
- Verify that initially there are 0 Functions
- Create the SQL-based UDF




```
CREATE FUNCTION f_concat_ws(varchar, varchar)
RETURNS varchar
STABLE
AS $$
  SELECT CONCAT($1, CONCAT(' ', $2))
$$ LANGUAGE SQL;
```

- Click **Run all** to execute the function creation query
- Refresh the Functions section
- Verify that the function f_concat_ws is visible
- Query the users table using the UDF




```
SELECT
    f_concat_ws(firstname, lastname),
    username
FROM users;
```

- Verify that the query returns:
  - Username
  - Full name with a space between first and last names
- Confirm that the UDF successfully concatenated user names in the expected format like John Smith